

ENG151

Science & technology

1st midterm

Notes

Page

Science: effort to produce empirical, theoretical, and practical knowledge about nature through observation, explanation, and prediction.

Technology: Using scientific knowledge in practical life

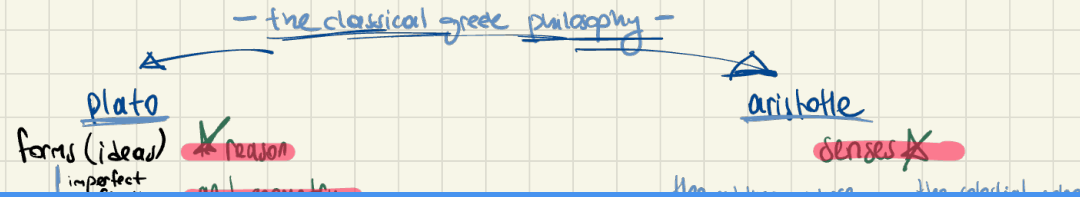
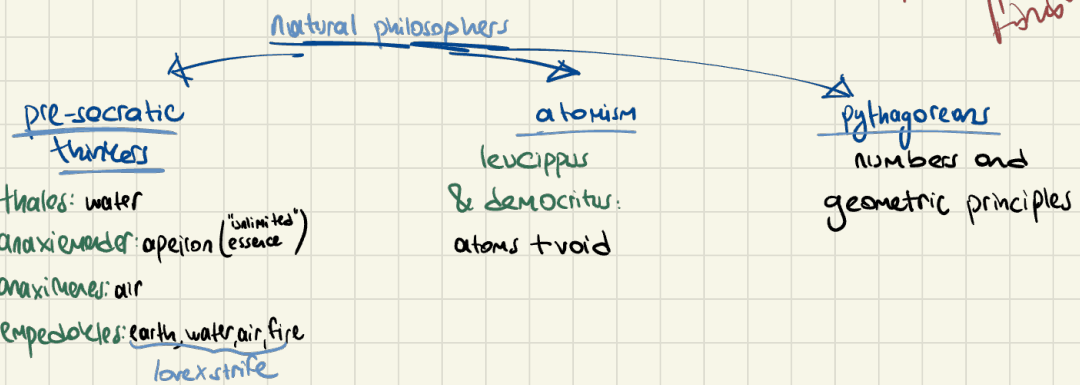
ancient times } by product of struggle for survival and order

Mesopotamia

- babylonian astronomy, sexagesimal
 - ↳ movements of celestial bodies
- āsipu, asu
- plimton 322
 - ↳ Pythagorean triples before the theorem.

transmission from philosophy to science ✕
Natural philosophy

ancient greece & hellenistic period



apprenticeship
first scientific journal
philosophical transaction of royal society
1665
periodical all kinds of articles

Egypt

- geometry (flooding of the Nile)
- anatomy (mummification)
- neuroscience (The Edwin Smith papyrus)
- modern scientific method (Ebers papyrus)

scientific revolution ✕
~~revolution~~
+ de revolutionibus of copernicus
+ principia of newton
mechanical motion

the hellenistic period

alexandria: the library of Alexandria and the museum

euclid: geometry, "Elements"

archimedes: statics (lever), hydrostatics, π , modern calculus

eratosthenes: Circumference of earth and the tilt of its axis

the antikythera mechanism: analog comp. that predicted eclipses and planets' movements

✱ zenith of engineering

(✱ collapse of Roman Empire →)

middle ages



byzantine

intense exchange

islamic

✱ university of constantinople

✱ copying and preserving ancient greek's work

al-khwarizmi: algebra, hindo-arabic numeral system to ✱ west

omer khayyam: algebra, cubic equations

al-rashi: decimal fractions

blugh beg: observations ahead of his time (such as star catalogs)

al-razi: smallpox, measles

al-beruni: ibn khayyam: Modern chemistry (sulfuric, nitric acid lab techniques)

ibn al-haytham: optics

al-kindi: optics

Nicolaus copernicus:

✱ on the revolutions of the heavenly spheres (1543)

✱ heliocentric model

galileo galilei: father of observational astronomy, modern physics, scientific method, modern science

✱ telescope

✱ jupiter's moons, phases of venus

✱ house arrest for life.

human body & scientific method

✱ andreas vesalius ✱ francis bacon

✱ william harvey ✱ robert hook micrographia

reason & mathematics

✱ rene de descartes

✱ robert boyle (PXV)

public verification of science } royal society of London

Isaac Newton:

✱ mathematical principles of natural philosophy

✱ write the works before him

✱ modern calculus

✱ optics ✱ binomial thm. galileo + copernicus

✱ paved the way for

✱ toledo school of translators / circulate again in the West

Medicine

✱ Avicenna
(the canon of medicine)
✱ Mozes

Maths & astronomy

✱ Ptolemy's almagest
✱ al-khwarizmi
✱ Euclid

Philosophy & natural science

✱ Aristotle
physics, on the heavens,
on generation and corruption,
nicomachean ethics.

Robert Grosseteste, Roger Bacon: empirical method.
William of Ockham, Ockham's razor: principle of parsimony ("simplicity")
Jean Buridan, the impetus theory: impetus, motion
the Oxford calculators: kinematics of motion

the black death

→ HALT!

✱ fall of Constantinople (1453)

printing process

scientific revolution

aristotelian natural philo. to chem. and biological sciences

philolaus, barut, montana

Kepler, Francis Bacon, more

✱ Descartes

✱ microscopy

planetary motion
✱ Scientific Method

ptolemy geocentric

X copernicus heliocentric

(andreas vesalius
✱ cadaver (galen))

galvani, allessandro volta
✱ muscle contraction
✱ battery

Laid the foundations of today's societies!

Age of Enlightenment (18th century)

reason & science ✱

+ social &

philosophical domains
(Newton, Descartes, Leibniz)

✱ Encyclopedie ✱ (dennis diderot)

↳ banned by french government

✱ Kant's "what is Enlightenment?" essay



The Industrial Revolution: A Quick Summary

What was it?

It was a time (starting in the late 1700s) when people stopped making things by hand and started using machines and factories.

Where did it start?

It started in Great Britain.

Why did it happen (Why did it start)?

The main reason was the invention of new technology, like the steam engine.

Also, Britain had three important things:

Fuel: Lots of coal to power the new machines.

Materials: Plenty of iron to build the machines and factories.

Workers: Many people who moved from farms to work in the new factories.

These factors together made it easy and profitable to build factories and produce goods very quickly.



Constantinople's Legacy and the Enlightenment

The Role of the Invasion (1204) & Conquest (1453)

The Latin invasions of Constantinople, especially the final Ottoman conquest in 1453, had an unintended positive effect on Western European knowledge.

How it worked:

The Treasure: For centuries, Constantinople kept the most important ancient Greek science and philosophy texts safe in its libraries (like the University of Constantinople).

The Escape: When the city was attacked and finally conquered (1453), many Byzantine scholars fled to Italy, taking these precious classical manuscripts with them.

The Awakening (Renaissance): These texts introduced Western Europe to forgotten rational thinking and human-centered ideas from antiquity, starting the Renaissance.

The Result (Enlightenment): The culture of reason and inquiry sparked by the Renaissance later developed into the Age of Enlightenment (1700s).

In one sentence:

The fall of Constantinople forced ancient knowledge to be moved West, which helped create the intellectual foundation for modern rational thought and the Enlightenment.